

student data management program

1. **File Management:** Create, read, write, and append to JSON files.
2. **JSON Data Handling:** Load, dump, and parse JSON data.
 - a. Assume in both file record of student and teacher there is no duplication of name in same file.
 - b. new data should be recorded in new line to increase readability.

```
[{"name": "hari", "email": "hari@gmail.com", "phone_number":  
{"name": "sita", "email": "sita@gmail.com", "phone_number": :  
{"name": "rita", "email": "rita@gmail.com", "phone_number": :
```

3. **Class Definition:** Define Teacher and Student class.

- a. **Teacher attributes**

- i. **Attributes** (order of attribute doesnot matter)

1. name, subject, id, address
2. email (check validation of the email syntax)
3. phone_number (dtype = int, len = 10), etc

- ii. **Methods**

1. accept (get info from user and store then in file)
2. display_all (Display general public information only like (name, email, phone_number, subject)
 - a. eg of student record

```
Name : hari
Email Address : hari@gmail.com
Phone number : 1425362514

Name : sita
Email Address : sita@gmail.com
Phone number : 1231231231

Name : rita
Email Address : rita@gmail.com
Phone number : 1425362514

Name : gita
Email Address : gita@gmail.com
Phone number : 1231231231
```

3. search (Display full detail of the requested person)
 - a. eg of data of student record

```
Enter the name of the Student: sita
Name : sita
Email Address : sita@gmail.com
Phone number : 1425142514
Roll number : 21
Marks : {'c': 54.0, 'c++': 14.0, 'python': 65.0}
Percentage: 44.333333333333336
```

4. delete (delete the record of particular name)

b. Student

i. attributs

1. name, roll_number, email, phone_number, marks, address

ii. **Methods**

1. accept, display_all, search (use abstraction for student)

2. **calculations**

- a. Pass_Fail_Determination():
- b. Highest_and_Lowest_Scores():
- c. Percentage():
- d. Rank_Calculation():

3. eg:

```
def search():  
    # display others information of the person such as  
    Pass_Fail_Determination():  
    Highest_and_Lowest_Scores():  
    # perform below actions if student is pass  
    Percentage():  
    Rank_Calculation():
```

4. **Authentication:**

- a. no authentication for first entry
- b. only teacher can add new data so Authenticate valid teachers using name and id.

```
print("You need to verify yourself as a teacher to add new entry")  
name = input("Enter your name: ")  
id = input("Enter your Id number: ")
```

5. **Data Validation:** new entry for student must have unique roll number and unique id in case of teacher

6. **Exception Handling:** Define and handle custom exceptions
(NoMatchingNameError, NoMatchingIdError, AuthenticationError).